Robot Procedure 2016_025

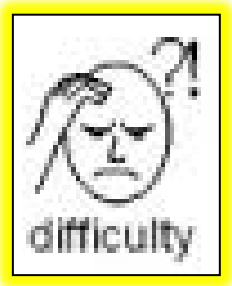
Jun 30th

2016

Procedure for Setting up log Diagnosis data in Robot Controller

[Applicable for all Robot Controller Model R30iA and later series]

If an unidentified trouble occurred...
Don't running ROBOT.
Don't POWER off.



How to SAVE DIAGNOSTIC Log!!

Diagnostic log

- Diagnostic log files used to store the internal data while some problems occur.
- To get necessary data with easy operation while occurring some problems and utilize for investigation.
- It will be realized early resolution of problems and reduction of investigating time.
- Some necessary data to investigate the problems are deleted after power off.
- This function also enables to save these data to FROM.
- This item also saves the log data to external device when problem occur on controller. This save operation should do before controller power off.

Limitation:

- Depending on the condition of occurring problem, this function is not available. For example, in the
- case TP is frozen and any key operation are ignored.
- Data is not saved when there is not enough empty capacity in FROM. Then warning, "Lack of FRA for saving", is displayed.
- If you execute this function again, data is overwritten. Diagnostic log is saved only 1 time of recent alarm occurred.
- For detail about this function, refer "DIAGNOSTIC LOG GETTING FUNCTION" in Robot Controller maintenance manual. This function can be used in the following software in R30iA controller:
    - Series 7DA1 version 45 or later
    - Series 7DA3 version 35 or later
    - Series 7DA4 version 19 or later
    - Series 7DA5 version 09 or later
- In case of R30iB this function is standard in all series of softwares

File formate:

- Diagnostic files (*.DG) used to store the specific diagnostic data in ASCII format.
- Usually Diagnostic (research) data can be saved when memory card has over 80MB vacant space.
- However in the case of a lot of files are existed in it, data cannot be saved normally. Then we recommend to use empty memory card (>128MB).

Format the Memory Card (MC:) or USB (UD1) with the Robot Controller

1. Prepare flash ATA card or USB that is >128MB.
2. Insert it to the robot controller at MC holder or USB
3. Press menu key to display the menu on teach pendant.
4. Select "7. FILE". FILE screen is displayed.
5. Press F5[UTIL] key and select "Set Device".
6. Select "Mem Card (MC:)" or “(UD1:)”
7. Press F5[UTIL] key and select "Format".
8. "Format disk?" is displayed. Select F4[YES].
9. "Enter volume label :" is displayed. Press ENTER key.

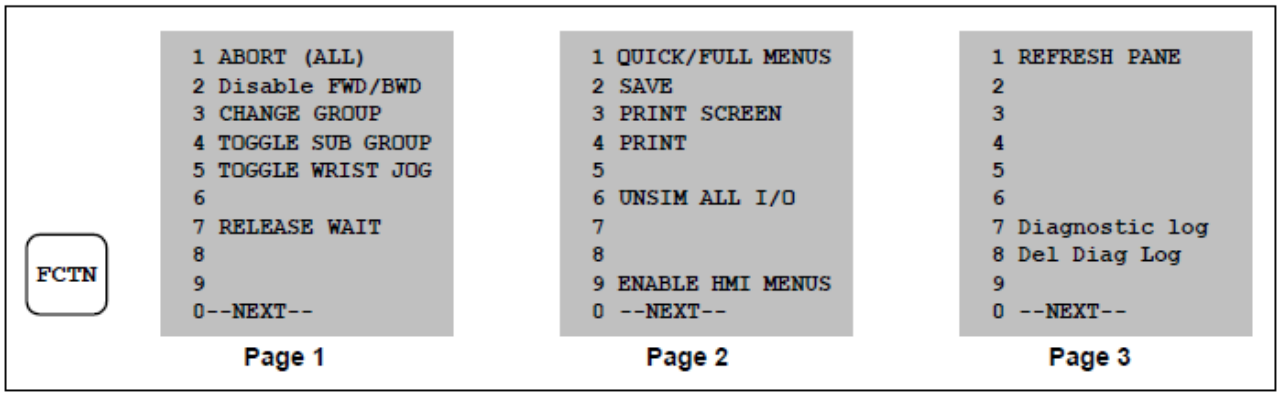
Save Diagnostic Log to MC or USB:

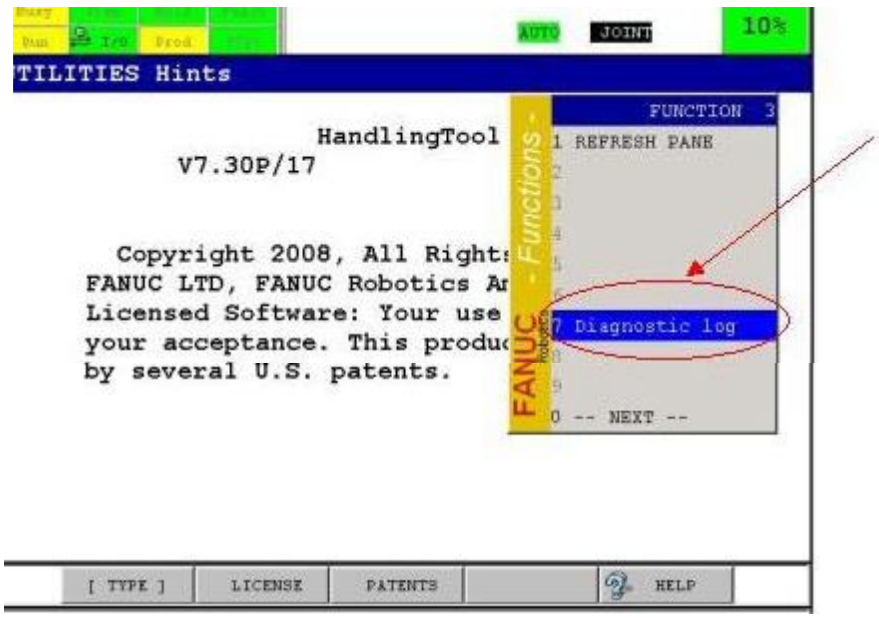
WARNING!

Confirm the robot is stopped due to some alarm!!

- If diagnostic log is saved while the robot is running, it gets jerky motion while saving the log in some cases.

1. Select "7. FILE". FILE screen is displayed.
2. Press F5[UTIL] key and select "Set Device".
3. Select "Mem Card (MC:)" or “(UD1:)”
4. Press FCTN key. Select " 0 -- NEXT -- " -> " 0 -- NEXT -- ", then "7 Diagnostic log" is shown on the third page.





5. Set the cursor onto "Diagnostic log" and press ENTER key. Diagnostic log is started.
6. While saving the log, "SYST 276 Now getting log..." is displayed on upper right of the teach pendant.
7. Maintenance data is saved into MNT_DATA¥ under the current directly. For instance, it is saved into MC:¥MNT_DATA¥ in the above case
8. It takes about one minutes to complete saving.
9. After saving is done, "SYST-274 Diagnostic log done" is displayed.
10. Please execute image backup after doing diagnostic log. Image backup allows the diagnostic log data to save external device such as memory card (MC).

Execute Image Backup

Refer to the Robot Maintenance manual to know the detail of image backup.

1. Insert the memory card to the controller.
2. Set the teach pendant enable switch to ON.
3. Select Menu – 7. FILE – F4[BACKUP] – Image backup.
4. [Destination Device] is displayed. Select Memory card (MC:).
5. Select F4[YES] after confirm message for cycle power is displayed.
6. Please cycle power. Image backup is started automatically.
7. Please do not push any key until done the image backup.
8. “Image backup completed successfully.” is displayed after done the image backup.
9. Select F4[OK], now controller start booting in normal mode

Know save all the collected data to one folder along with information of Robot & Controller name plate and send it across to us for diagnosing the possible reasons for intermittent and permanent alarms.